

Advanced React Masterclass – Module 3 Exercise

Hands-on Exercise No. 3

Course: Advanced React Masterclass

Module: Performance Optimization

Total Marks: 10

Due Date: 16/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Apply memoization techniques in React
 - Use React.memo to prevent unnecessary re-renders
 - Optimize performance using useMemo and useCallback
 - Implement lazy loading for components
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Problem Statement:

You are working as a Frontend Engineer for a dashboard application. The client is facing performance issues when rendering large datasets. Your task is to optimize the React application using memoization techniques and lazy loading.

Task Requirements:

- Create a React project with a Dashboard component displaying a list of 1000 items
- Use React.memo to prevent unnecessary re-renders of child components
- Apply useMemo to cache expensive calculations (such as filtering items)
- Use useCallback to memoize event handler functions
- Implement lazy loading for a chart component that loads only when required
- Take screenshots of:
 - Code using React.memo, useMemo, and useCallback

- Browser output before optimization
 - Browser output after optimization
 - Lazy loading implementation
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Code snippets
 - Screenshots of performance improvements
 - Output comparisons
- Add short explanations for each optimization technique